

Paper Summary for Prof. Xin Yuan

Physics World Models for Computational Imaging:
CASSI, CACTI, and SPC Validation Results

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February 2026 — Prepared for Nature submission

Paper Overview

This paper introduces **Physics World Models (PWM)**, a universal diagnostic and correction framework built on the **Triad Decomposition**, which attributes imaging failures to three gates: recoverability (Gate 1), carrier budget (Gate 2), and operator mismatch (Gate 3). The **OperatorGraph IR** encodes forward models across modalities into a common DAG formalism, enabling modality-agnostic diagnosis.

Central finding. Gate 3 (operator mismatch) dominates in *every* validated modality. A sub-pixel mask perturbation erases **twice the reconstruction gains** of a decade of solver innovation.

Sections Most Relevant to Your Expertise

1. CASSI Deep Dive: GAP-TV, MST-L, HDNet Under Mismatch

The paper uses the 5-parameter InverseNet mismatch ($dx=0.5$ px, $dy=0.3$ px, $\theta=0.1^\circ$, $a_1=2.02$, $\alpha=0.15^\circ$) across 10 KAIST scenes:

Method	Sc. I	Sc. II	Sc. IV	ρ
GAP-TV	24.34	20.96	21.72	0.22
MST-L	34.81	20.83	27.33	0.46
HDNet	34.66	21.88	21.88	0.00

Key insight: All solvers collapse to a narrow Scenario II range (20.83–21.88 dB), confirming the failure is operator-driven. GAP-TV achieves $\rho = 0.22$ recovery; MST-L achieves $\rho = 0.46$ because it conditions on the mask. HDNet’s mask-oblivious architecture shows $\rho = 0$ —it cannot exploit a corrected mask.

2. CACTI Results: EfficientSCI and Temporal Compression

CACTI exhibits the most severe mismatch sensitivity among photon-domain modalities:

Method	Sc. I	Sc. II	Sc. III	ρ
GAP-TV	26.75	15.81	26.01	0.93
PnP-FFDNet	29.28	11.43	25.39	0.78
ELP-Unfolding	34.09	15.47	29.40	0.75
EfficientSCI	35.39	14.81	27.38	0.61

Key insights: All four methods suffer catastrophic mismatch degradation (-10.94 to -20.58 dB), confirming operator-driven failure. GAP-TV achieves the highest recovery ratio ($\rho = 0.93$), recovering $+10.21$ dB of the 10.94 dB degradation—nearly perfect calibration. EfficientSCI has the lowest ρ (0.61) despite the best Sc. I performance, indicating that deep unrolling networks suffer more *irreversible* damage from mismatch than iterative methods. PnP-FFDNet and ELP-Unfolding fall in between ($\rho = 0.78$ and 0.75). The temporal mask replication amplifies calibration errors multiplicatively across all compressed frames.

3. SPC Gain Mismatch: A Qualitatively Different Failure Mode

SPC presents **gain bias** rather than geometric shift:

- 5% detector gain drift $\Rightarrow -12.21$ dB PSNR drop
- PWM diagnoses Gate 3 localized to detector gain node
- Correction recovers the full 12.21 dB ($\rho = 1.0$)
- **But:** Gain-type mismatch is *not* recoverable from measurement residual alone (Scenario IV recovery = 0% for SPC), because gain bias scales signal and noise uniformly

4. Real-Data Hardware Validation

CASSI real (TSA dataset, 5 scenes):

- GAP-TV residual ratio: $1.8\times$ (consistent across 5 scenes, range $1.6\text{--}2.0\times$)
- HDNet: $1.0\times$ (mask-oblivious, insensitive)
- MST-S/MST-L: $\sim 1.0\times$ on real data (vs. -13.98 dB in simulation!)
- **Explanation:** Real hardware mask contains pre-existing manufacturing errors; additional 0.5 px perturbation is small relative to existing imperfections

CACTI real (4 scenes):

- GAP-TV residual ratio: $10.4\times$ (range $9.4\text{--}11.0\times$)
- PnP-FFDNet: $2.0\times$ (deep prior provides partial robustness)
- Temporal compression amplifies errors across all 10 frames

Autonomous calibration:

- CASSI: Grid search recovers 78% of oracle correction (1,234 s)
- CACTI: Grid search recovers **100%** of oracle correction (64 s)
- SPC: 0% recovery (gain mismatch invisible to residual-based calibration)

5. SSIM and SAM Supplementary Results

CASSI SAM	Sc. I	Sc. II	Sc. III
GAP-TV	16.66°	24.27°	25.97°
MST-L	7.44°	23.92°	11.74°
HDNet	6.67°	17.03°	17.03°

MST-L’s SAM increases from 7.44° to 23.92° under mismatch—substantial spectral mixing. Oracle correction partially recovers to 11.74°.

Your Role and Contributions

As **co-author**, your contributions include:

1. Domain expertise on CASSI and CACTI forward models and the mismatch parameter specifications used in all experiments
2. Validation of GAP-TV and EfficientSCI reconstruction algorithms
3. Expertise on SPC gain-mismatch failure mode (qualitatively different from geometric mismatch)
4. Manuscript editing and domain-specific review

Paper Statistics

- Main paper: 34 pages, 7 figures; Supplementary: 22 pages, 12 tables
- 10 KAIST scenes (CASSI), 6 benchmark videos (CACTI), 11 SPC images
- 5 real CASSI scenes + 4 real CACTI scenes
- Correction gains: CASSI +0.76–6.50 dB, CACTI +22.94 dB, SPC +12.21 dB